

Scattered spring: How Warming Climates Shift Ecosystem into Higher Variability

Jonathan Auerbach, Andrew Gelman & E.M. Wolkovich

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1 Outline: 30 September 2025

1. Introduction

- (a) Climate change is shifting biological systems globally with potential impacts on ecosystem stability and the global climate system itself [1]
 - i. Shifted phenology—the timing of recurring life history events—has been the clearest ‘fingerprint’ of anthropogenic climate change across both natural and managed systems
 - ii. ... with major impacts....
 - iii. Spring frosts are increasing, with major impacts on forests [?] and major crops (e.g., winegrapes)
 - iv. Additionally, there are increasing reports of ‘abnormal seasonal activities’ (<https://www.nature.com/articles/s41598-025-16585-2>)
 - v. Uneven shifts across some trophic levels has driven concern of declining pollination and other ecosystem services with continued warming
 - vi. Most concerns of trophic mismatch premise that different trophic levels use different cues to time life history events and thus may become desynchronized with continued warming, but synchrony within a species or population is also often critical
- (b) Here we show how warming may be desynchronizing spring plant phenology (flower and leaf-burst, ‘budburst’) without any shift or difference in the cues.
 - i. Using the simplest model—a thermal sum accumulation model, which is underlying model for many phenological events across plants and insects—we show how the pattern of temperatures predicts increasing variance with continued warming
 - ii. Our main finding is that the linear increase of average temperatures, which is definitional to climatological ‘spring’ months makes plants budburst at the same time, with variation between individuals negligible. But as temperatures warm, budburst happens in the more stationary temperature regimes (where average temperatures

do not increase) and small variations have a larger effect, resulting in the desynchronization of blooms or a scattered spring.

- (c) by shifting phenological triggers (thresholds?) earlier into more stationary seasonal climate periods

2. Explaining spring versus winter ...

- (a) This theory suggests the synchrony between buds depends critically on whether temperatures accumulate predominately in relatively flat or stationary weather periods or during periods of day over day warming periods.
- (b) Such static or stationary weather regimes generally define the traditional winter (January - March, and summer months: June-August) while warming defines the spring (March - May), though shorter periods of either regime type can happen generally in any month

3. (Results &) Discussion (just discussion for now)

- (a) Here's just an dis-organized list of stuff to cover ..
 - i. Experiments (always done at constant or at most diurnally fluctuating temperatures) show predicted higher variance at lower temperatures (**higher than predicted, right?** Because all the temperature treatments – how to high – are applied as constant temperatures ... but I think we're talking about variance that is more than predicted by the simple thermal model so we need to spell this out for folks)
 - ii. Plant may take advantage of this mathematical relationship to increase within-population synchrony each year and thus promote effective reproduction (super-synchrony? super-consistency?)
 - iii. This finding should impact studies and predictions phenological fitness optima, including when and where co and counter-gradient variation are favored, which has (to date) focused on (which type? Constant or linearly increasing) of temperature variation [?]
 - iv. Something about how – how long the stationary vs linear increasing period needs to be for variance to change depends on the base temperature for GDD and also the total sum of GDD needed
 - v. We don't know for sure what causes all variation in phenology between individuals but microclimate plays a major role ... our results show that linearly increasing spring temperatures make effects of microclimate variation negligible, resulting in buds with near simultaneous blooms, but this is not true during periods of stationary temperature periods....
 - vi. Implications for crop forecasting ...
 - vii. Implications for forests – maybe more frosts? Definitely should affect gene flow.

2 Outline: 26 August 2025 – tipping point

1. Introduction

- (a) Climate change is shifting biological systems globally with potential impacts on ecosystem stability and the global climate system itself [1]
 - i. The most observed and well documented biological impact is shifts in phenology—the timing of recurring life history events [2, 3, 4]
 - ii. Shifted phenology has been the clearest ‘fingerprint’ of anthropogenic climate change, it may also be the one to first show evidence that systems are no longer responding to climate change as they did in the past [5, 6]
 - iii. Such a phase change or tipping point [7] have been seen in other human-altered systems (but understanding why they happen and predicting them has been difficult) [8]
 - iv. Understanding and predicting this has big consequences for global carbon models (will plants sequester more carbon?), economic models of crops (peach frost etc.) and more... [9, 10, 11]
- (b) Research is divided on whether phenological responses are showing hints of new responses [12, 13] or not [14], which may because we lack an underlying model
 - i. Declining sensitivities, or not
 - ii. Increasing variance, or not
 - iii. Related to this, are trees growing more with longer seasons, or not (or maybe we just focus on phenology) [15, 16]
 - iv. These debates stem from lacking an underlying model that we all agree on (and we don’t have great data)
- (c) Here we show how shifted autocorrelation in climate (or do we want to call it ‘biological climate’ or something to represent that it’s the GDD metric? Though I am not a fan of new names/terms) due to anthropogenic climate change predicts a breakdown of one of the most fundamental models of plant biology and show how it may lead to a decline in forecasting accuracy
- (d) We show results with lilacs
- (e) And with other data
- (f) Our results suggest climate change has fundamentally disrupted the central limit theory for one of the most important biological models with consequences for higher variance and lower predictability from crops to forests and likely extends to upper trophic levels as well.

2. Results & Discussion

- (a) We found ...
 - (b) Changes in autocorrelation in climate already observed (maybe muse on shifts in min versus max, vs mean)
3. Methods
- (a) Longer form of the model
 - (b) Methods for lilac data
 - (c) Methods for other datasets (PEP725, Oak in UK, cherry blossoms)

Stuff I think we should try to fit in somewhere ...

1. Ecologists are really into increased environmental variation (since it does cool stuff in our deterministic models of the world), but we show that decreasing environmental variation – increases in autocorrelation – could actually be what is happening (and it's more interesting) (**author?**) [17, 18, 19]

Some thoughts of stuff that could be related to shifting variance as we suggest here ...

1. More wide-spread frost events (because plants all budburst over larger spatial areas at the same time)

Stuff worth checking out:

- Paper on variance and phenology ... Disorder or a new order: How climate change affects phenological variability
 - Autocorrelation (on a different temporal scale I assume) used to help identify human-caused climate change Anthropogenic Influence on the Autocorrelation Structure of Hemispheric-Mean Temperatures and replies.
 - Increased spatial and temporal autocorrelation of temperature under climate change
- 1.

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3 Figures

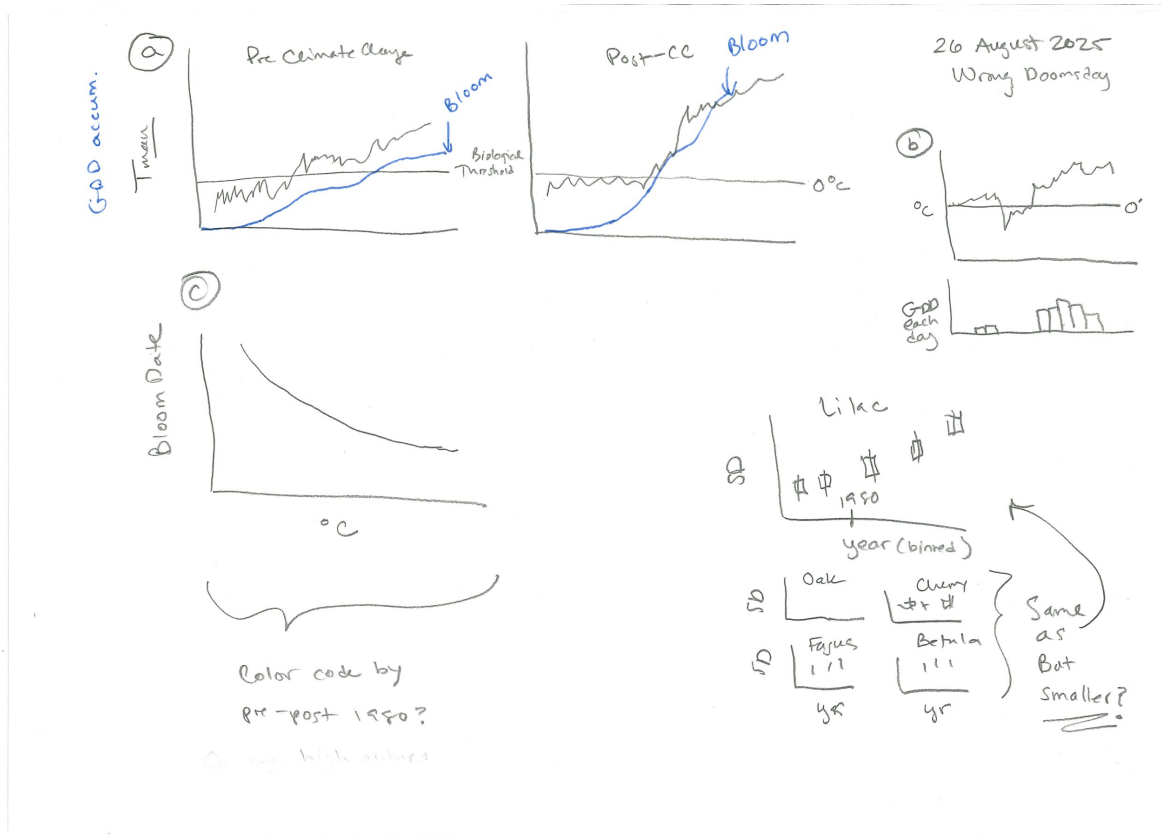


Figure 1: Ideas for figures: I started with a) as a conceptual figure to explain the problem but it's not great, we may need something like b) which would show climate, then daily GDD units and then the sum somehow. Some version of c) seems useful I think? I think we'll have to somehow show the trends in leafout over time and then I forgot to label the lower right bit that would be the main results. I am just riffing (and open to all ideas, obviously).